DVWA Installation and Configuration

Objective

Install and configure Damn Vulnerable Web Application (DVWA) on a local Kali Linux server to practice web application security testing in a controlled environment.

Reference Video

DVWA Installation Tutorial

Tools and Requirements

- Kali Linux
- Apache2 web server
- MariaDB database server
- PHP and required extensions
- Git
- DVWA

Installation Steps

1. Updated the package lists and installed Apache2, MariaDB, PHP, and required PHP extensions.
2. Started Apache2 and MariaDB.
3. Downloaded DVWA from the official GitHub repository.
4. Created the DVWA database and its database user.
5. Configured the database credentials in `config/config.inc.php`.
6. Opened the DVWA setup page in the browser.
7. Initialized the database using the Create / Reset Database button.
8. Logged in and selected the Low security level for initial testing.

Verification

- Apache2 service: [Pass/Fail]
- MariaDB service: [Pass/Fail]
- DVWA setup page accessible: [Pass/Fail]
- Database initialized successfully: [Pass/Fail]

- DVWA login successful: [Pass/Fail]

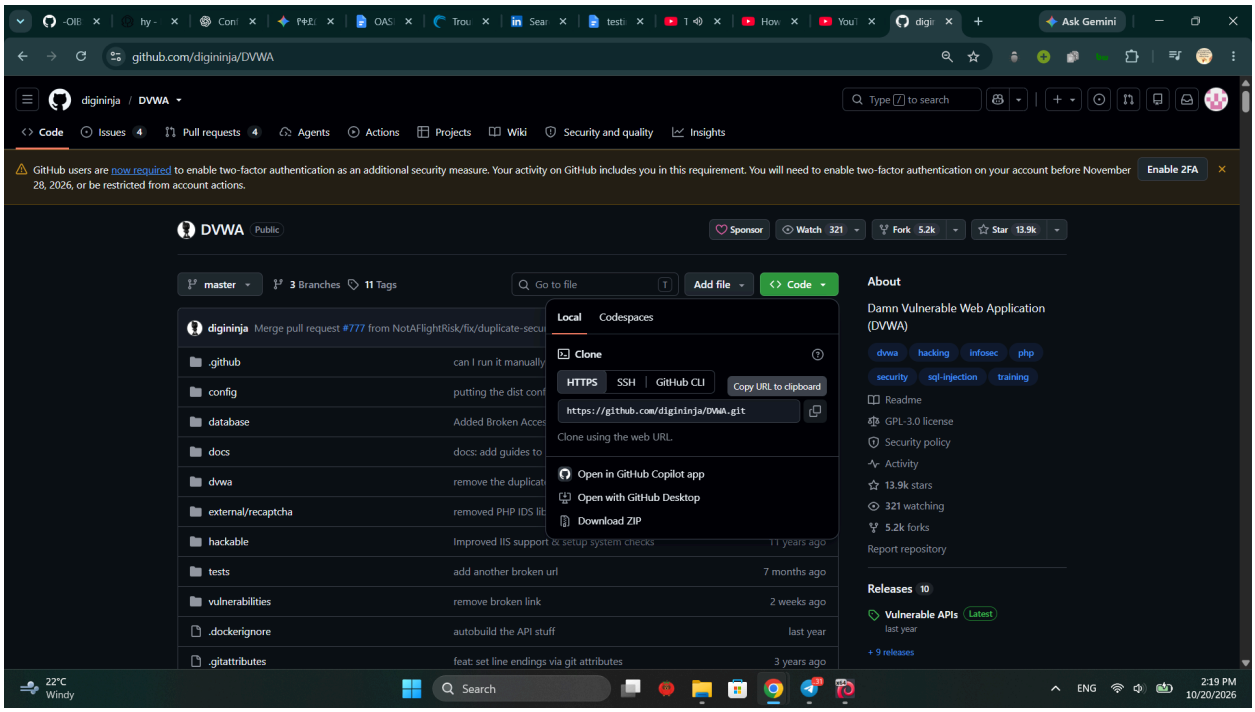
Local URL

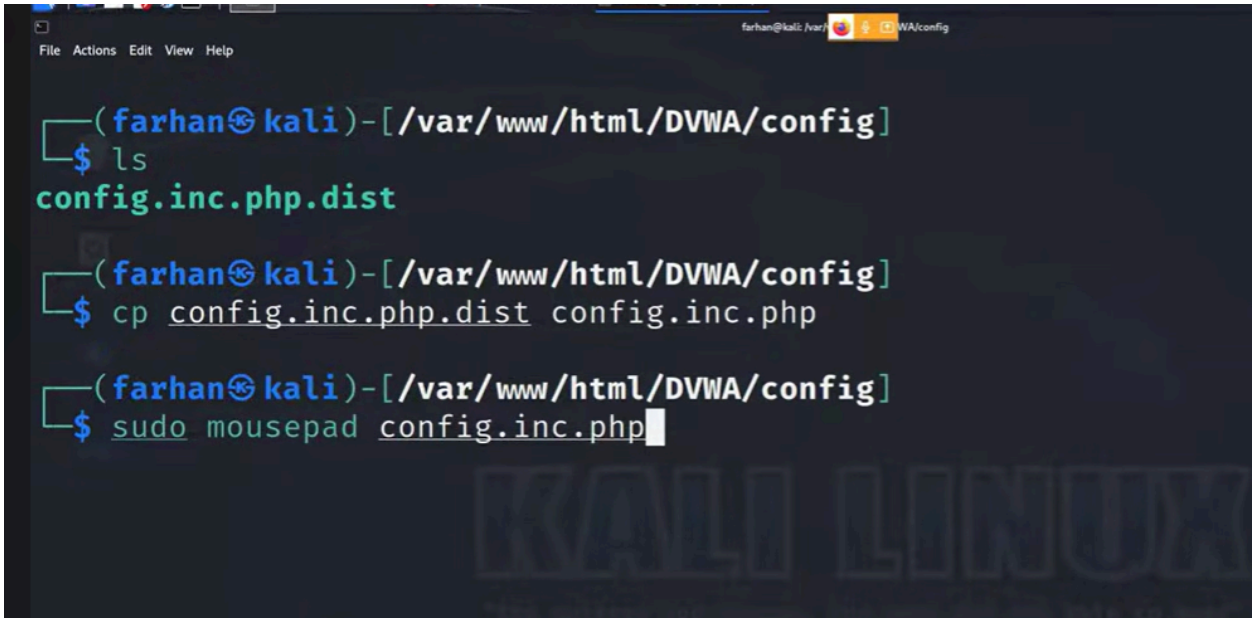
`http://127.0.0.1/DVWA/`

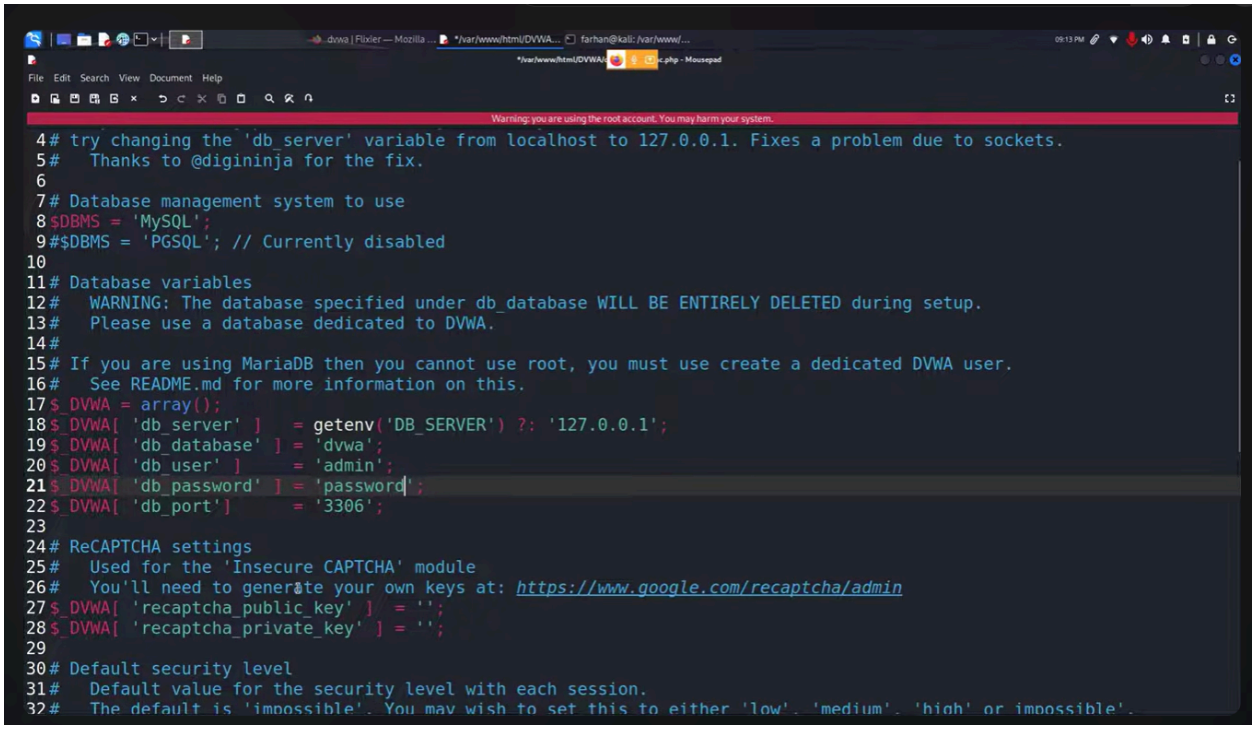
Security Note

DVWA intentionally contains vulnerable code. It must only be used in a local, isolated lab environment and must not be exposed to the public internet.

Result




File Actions Edit View Help
(farhan㉿kali)-[/var/www/html/DVWA/config]
$ ls
config.inc.php.dist
(farhan㉿kali)-[/var/www/html/DVWA/config]
$ cp config.inc.php.dist config.inc.php
(farhan㉿kali)-[/var/www/html/DVWA/config]
$ sudo mousepad config.inc.php
KALI LINUX


File Edit Search View Document Help
Warning: you are using the root account. You may harm your system.
try changing the 'db_server' variable from localhost to 127.0.0.1. Fixes a problem due to sockets.
Thanks to @digininja for the fix.
Database management system to use
$DBMS = 'MySQL';
#$DBMS = 'PGSQL'; // Currently disabled
Database variables
WARNING: The database specified under db_database WILL BE ENTIRELY DELETED during setup.
Please use a database dedicated to DVWA.
If you are using MariaDB then you cannot use root, you must use create a dedicated DVWA user.
See README.md for more information on this.
$_DVWA = array();
$_DVWA['db_server'] = getenv('DB_SERVER') ?: '127.0.0.1';
$_DVWA['db_database'] = 'dvwa';
$_DVWA['db_user'] = 'admin';
$_DVWA['db_password'] = 'password';
$_DVWA['db_port'] = '3306';
ReCAPTCHA settings
Used for the 'Insecure CAPTCHA' module
You'll need to generate your own keys at: https://www.google.com/recaptcha/admin
$_DVWA['recaptcha_public_key'] = '';
$_DVWA['recaptcha_private_key'] = '';
Default security level
Default value for the security level with each session.
The default is 'impossible'. You may wish to set this to either 'low', 'medium', 'high' or impossible'.

```
┌──(farhan㉿kali)-[/var/www/html/DVWA/config]
└─$ sudo mousepad config.inc.php
[sudo] password for farhan:

┌──(farhan㉿kali)-[/var/www/html/DVWA/config]
└─$ sudo systemctl start mysql
```

```
● mariadb.service - MariaDB 10.11.4 database server
     Loaded: loaded (/lib/systemd/system/mariadb.service; disabled; preset: disabled)
     Active: active (running) since Thu 2023-08-10 21:17:21 EDT; 8s ago
       Docs: man:mariadbd(8)
             https://mariadb.com/kb/en/library/systemd/
    Process: 4598 ExecStartPre=/usr/bin/install -m 755 -o mysql -g root -d /var/run/m>
    Process: 4599 ExecStartPre=/bin/sh -c systemctl unset-environment _WSREP_START_PO>
    Process: 4601 ExecStartPre=/bin/sh -c [ ! -e /usr/bin/galera_recovery ] && VAR= |>
    Process: 4678 ExecStartPost=/bin/sh -c systemctl unset-environment _WSREP_START_P>
    Process: 4680 ExecStartPost=/etc/mysql/debian-start (code=exited, status=0/SUCCES>
   Main PID: 4661 (mariadbd)
     Status: "Taking your SQL requests now..."
      Tasks: 24 (limit: 4442)
     Memory: 254.1M
        CPU: 629ms
     CGroup: /system.slice/mariadb.service
```

```
┌──(farhan㉿kali)-[/var/www/html/DVWA/config]
└─$ sudo su
┌──(root💀kali)-[/var/www/html/DVWA/config]
└─# mysql -u root -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 73
Server version: 10.11.4-MariaDB-1 Debian 12

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]>
```

```
Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> create database dvwa;
Query OK, 1 row affected (0.000 sec)

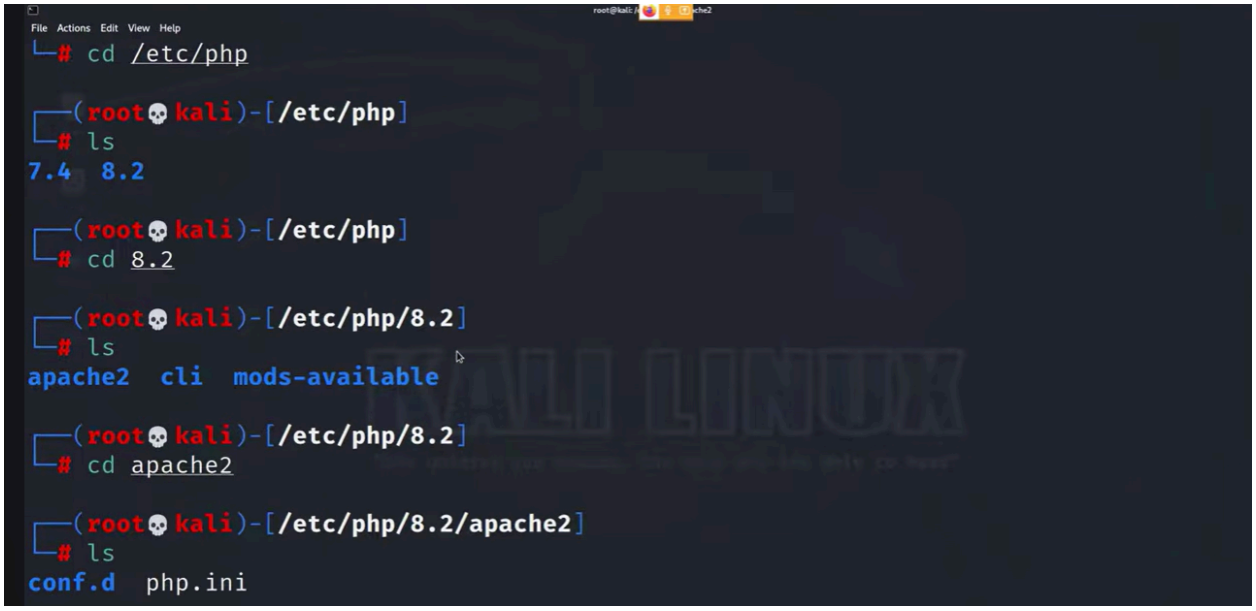
MariaDB [(none)]> create user 'admin'@'127.0.0.1' identified by 'password';
Query OK, 0 rows affected (0.033 sec)

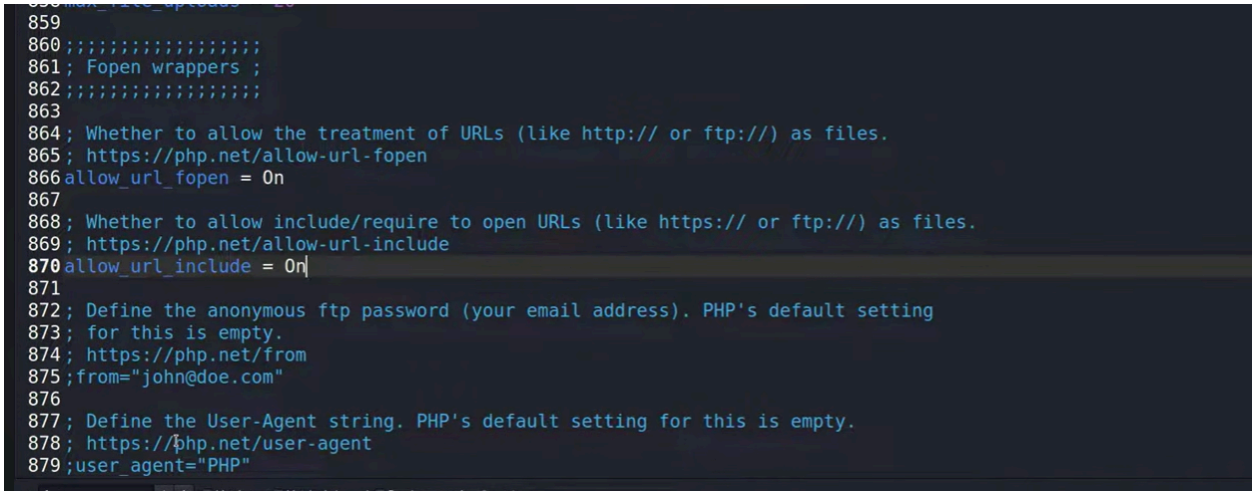
MariaDB [(none)]> grant all privileges on dvwa.* to 'admin'@'127.0.0.1';
```

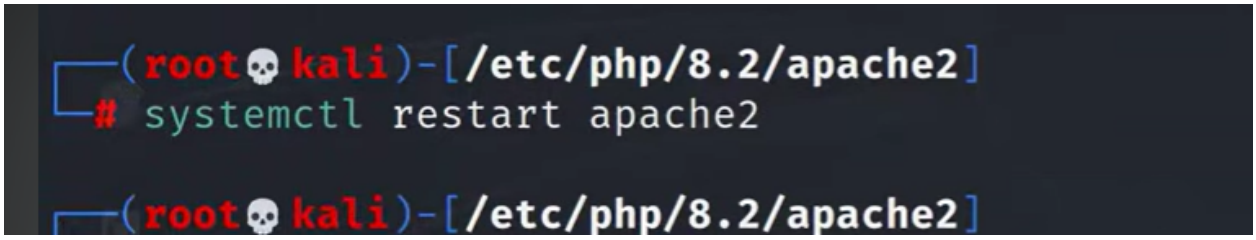
```
┌──(root💀kali)-[/var/www/html/DVWA/config]
└─# systemctl start apache2

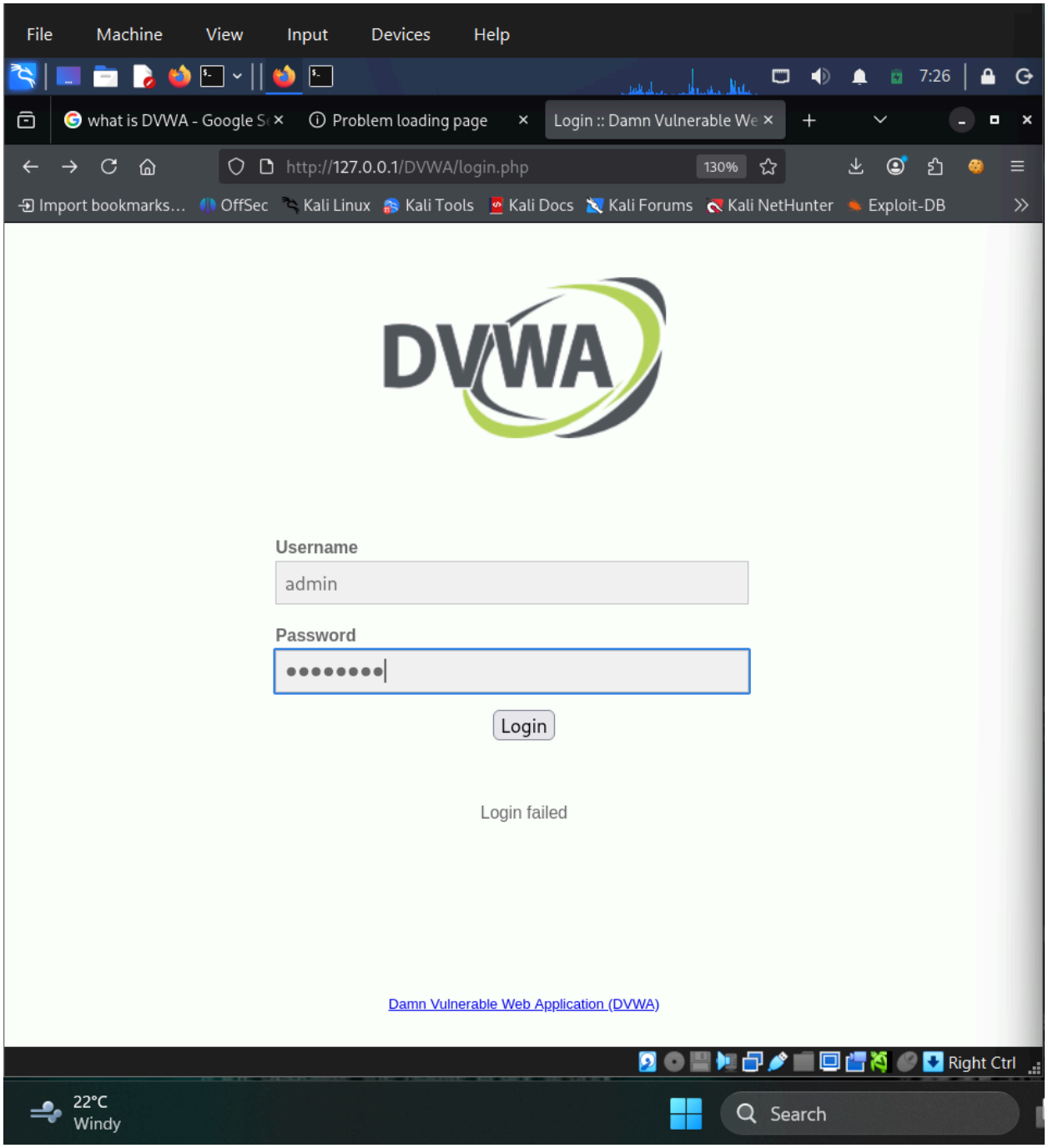
┌──(root💀kali)-[/var/www/html/DVWA/config]
└─#
```

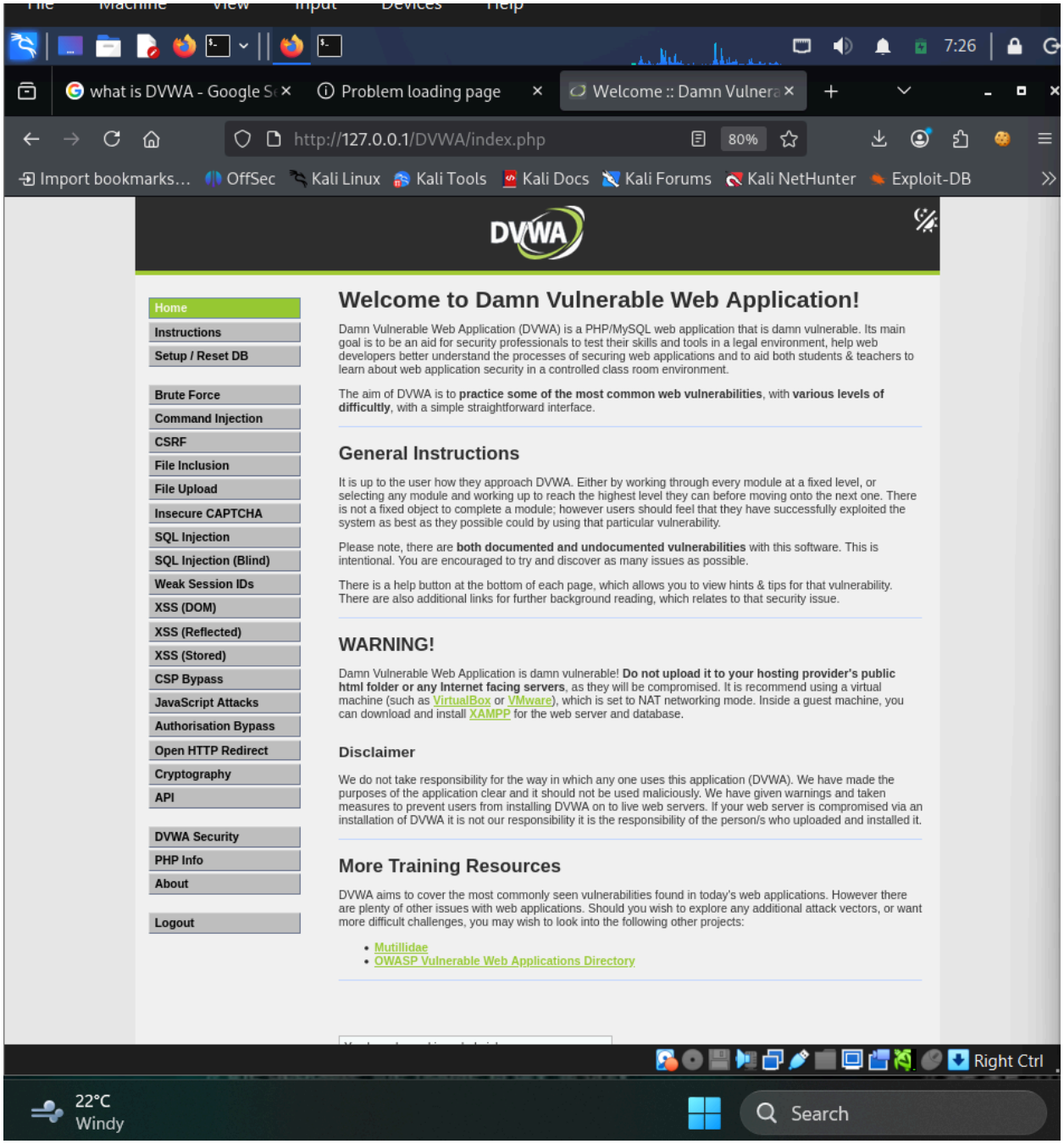
```
apache2.service - The Apache HTTP Server
     Loaded: loaded (/lib/systemd/system/apache2.service; disabled; preset: disabled)
     Active: active (running) since Thu 2023-08-10 22:06:24 EDT; 53s ago
       Docs: https://httpd.apache.org/docs/2.4/
    Process: 6283 ExecStart=/usr/sbin/apachectl start (code=exited, status=0/SUCCESS)
   Main PID: 6300 (apache2)
      Tasks: 6 (limit: 4442)
     Memory: 20.7M
        CPU: 164ms
     CGroup: /system.slice/apache2.service
             ├─6300 /usr/sbin/apache2 -k start
```


File Actions Edit View Help
cd /etc/php
(root kali)-[/etc/php]
ls
7.4 8.2
(root kali)-[/etc/php]
cd 8.2
(root kali)-[/etc/php/8.2]
ls
apache2 cli mods-available
(root kali)-[/etc/php/8.2]
cd apache2
(root kali)-[/etc/php/8.2/apache2]
ls
conf.d php.ini


859
860 ;;;;;;;;;;;;;;;;;;
861 ; Fopen wrappers ;
862 ;;;;;;;;;;;;;;;;;;
863
864 ; Whether to allow the treatment of URLs (like http:// or ftp://) as files.
865 ; https://php.net/allow-url-fopen
866 allow_url_fopen = On
867
868 ; Whether to allow include/require to open URLs (like https:// or ftp://) as files.
869 ; https://php.net/allow-url-include
870 allow_url_include = On
871
872 ; Define the anonymous ftp password (your email address). PHP's default setting
873 ; for this is empty.
874 ; https://php.net/from
875 ;from="john@doe.com"
876
877 ; Define the User-Agent string. PHP's default setting for this is empty.
878 ; https://php.net/user-agent
879 ;user_agent="PHP"


(root kali)-[/etc/php/8.2/apache2]
systemctl restart apache2
(root kali)-[/etc/php/8.2/apache2]


File
Machine
View
Input
Devices
Help
7:26
what is DVWA - Google S
Problem loading page
Login :: Damn Vulnerable We
http://127.0.0.1/DVWA/login.php
130%
Import bookmarks...
OffSec
Kali Linux
Kali Tools
Kali Docs
Kali Forums
Kali NetHunter
Exploit-DB
DVWA
Username
admin
Password
Login
Login failed
Damn Vulnerable Web Application (DVWA)
Right Ctrl
22°C
Windy
Search


File Machine View Input Devices Help
7:26
what is DVWA - Google S
Problem loading page
Welcome :: Damn Vulnera
http://127.0.0.1/DVWA/index.php
80%
Import bookmarks...
OffSec
Kali Linux
Kali Tools
Kali Docs
Kali Forums
Kali NetHunter
Exploit-DB
DVWA
Home
Instructions
Setup / Reset DB
Brute Force
Command Injection
CSRF
File Inclusion
File Upload
Insecure CAPTCHA
SQL Injection
SQL Injection (Blind)
Weak Session IDs
XSS (DOM)
XSS (Reflected)
XSS (Stored)
CSP Bypass
JavaScript Attacks
Authorisation Bypass
Open HTTP Redirect
Cryptography
API
DVWA Security
PHP Info
About
Logout
Welcome to Damn Vulnerable Web Application!
Damn Vulnerable Web Application (DVWA) is a PHP/MySQL web application that is damn vulnerable. Its main goal is to be an aid for security professionals to test their skills and tools in a legal environment, help web developers better understand the processes of securing web applications and to aid both students & teachers to learn about web application security in a controlled class room environment.
The aim of DVWA is to practice some of the most common web vulnerabilities, with various levels of difficultly, with a simple straightforward interface.
General Instructions
It is up to the user how they approach DVWA. Either by working through every module at a fixed level, or selecting any module and working up to reach the highest level they can before moving onto the next one. There is not a fixed object to complete a module; however users should feel that they have successfully exploited the system as best as they possible could by using that particular vulnerability.
Please note, there are both documented and undocumented vulnerabilities with this software. This is intentional. You are encouraged to try and discover as many issues as possible.
There is a help button at the bottom of each page, which allows you to view hints & tips for that vulnerability. There are also additional links for further background reading, which relates to that security issue.
WARNING!
Damn Vulnerable Web Application is damn vulnerable! Do not upload it to your hosting provider's public html folder or any Internet facing servers, as they will be compromised. It is recommend using a virtual machine (such as VirtualBox or VMware), which is set to NAT networking mode. Inside a guest machine, you can download and install XAMPP for the web server and database.
Disclaimer
We do not take responsibility for the way in which any one uses this application (DVWA). We have made the purposes of the application clear and it should not be used maliciously. We have given warnings and taken measures to prevent users from installing DVWA on to live web servers. If your web server is compromised via an installation of DVWA it is not our responsibility it is the responsibility of the person/s who uploaded and installed it.
More Training Resources
DVWA aims to cover the most commonly seen vulnerabilities found in today's web applications. However there are plenty of other issues with web applications. Should you wish to explore any additional attack vectors, or want more difficult challenges, you may wish to look into the following other projects:
Mutillidae
OWASP Vulnerable Web Applications Directory
Right Ctrl
22°C
Windy
Search